

# Preliminary Dense-RHS CoRL Compact Results

Autogenerated from local TD-MPC2-JAX run artifacts

**Scope.** This preliminary report summarizes the completed compact campaign “dense\_rhs\_corl\_compact\_v2” using local metric caches refreshed from the NCC run directories. It is intended as an internal CoRL-style evidence package, not as the final v3 matrix. The following environment is excluded from all results and plots: fish-swim.

**Headline.** Across 30 matched valid seed pairs, Adaptive RHS changes final score by **5.90%  $\pm$  24.00%** relative to the fixed TD-MPC2 paper horizon, with an RHS win rate of **70.00%**. The summary excludes 0 missing, nonfinite, or unmatched pairs.

## Experimental Setup

We compare the TD-MPC2 fixed paper horizon against Adaptive RHS under matched environment, regime, and seed cells. Environments are quadruped-run, cheetah-run, hopper-hop, finger-turn\_hard, cartpole-swingup; regimes are clean and chaos; seeds are 1, 7, 15; training budget is 500000 environment steps. All rows come from `experiments/corl_compact_v2_ledger.csv`; each row records the SLURM job id, git commit, remote run directory, score, runtime, and checkpoint status.

## Aggregate Result

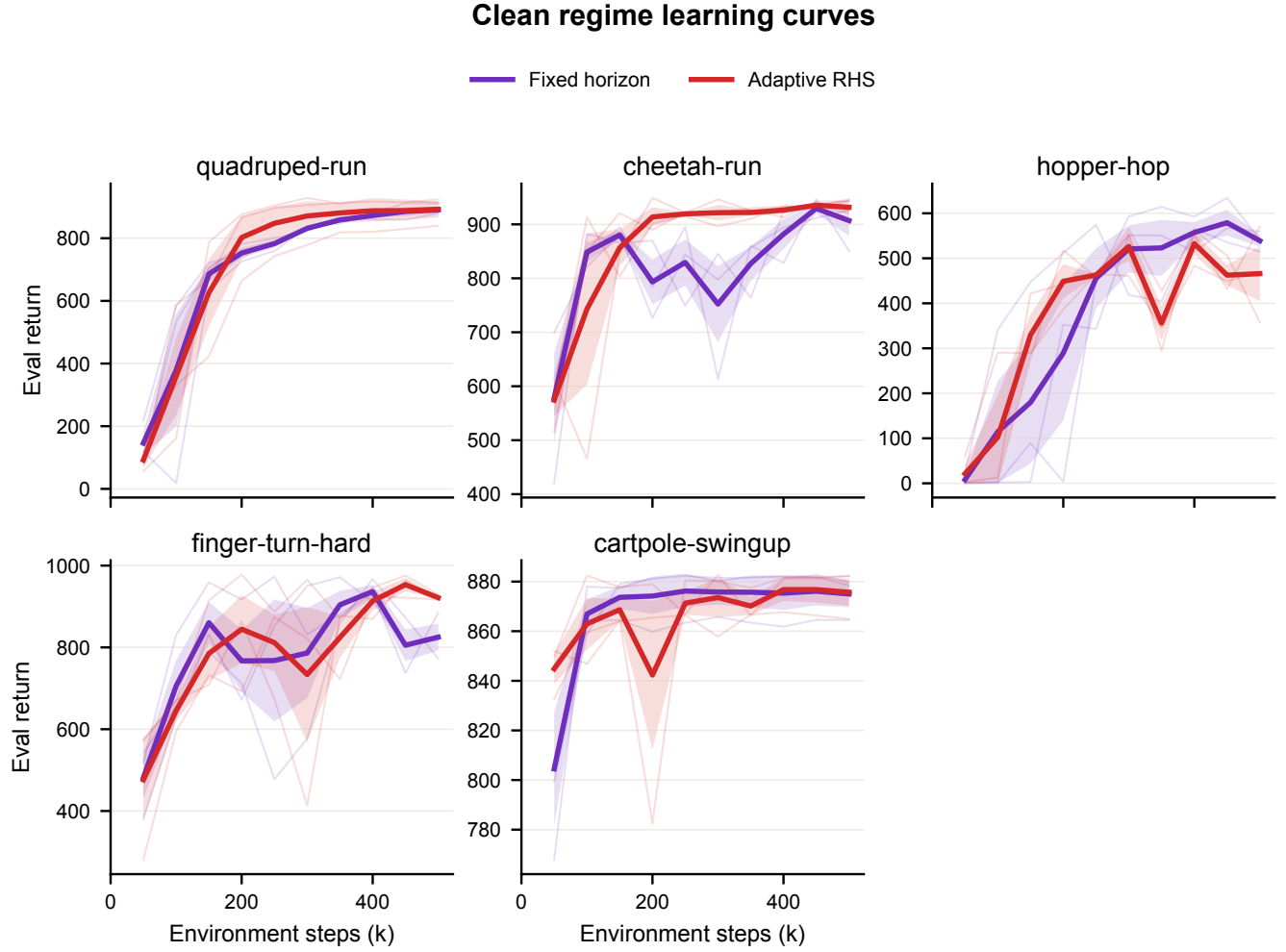
| Scope           | n  | RHS delta (%)     | RHS wins | Runtime ratio |
|-----------------|----|-------------------|----------|---------------|
| all valid pairs | 30 | 5.90 $\pm$ 24.00  | 70.00%   | 1.13          |
| chaos           | 15 | 11.43 $\pm$ 31.37 | 73.33%   | 1.13          |
| clean           | 15 | 0.37 $\pm$ 11.99  | 66.67%   | 1.13          |

## Full Matched Results

| Environment      | Regime | Seed | Fixed | RHS   | $\Delta$ (%) | Fixed h | RHS h |
|------------------|--------|------|-------|-------|--------------|---------|-------|
| cartpole-swingup | chaos  | 1    | 814.1 | 816.4 | +0.28        | 1.76    | 2.04  |
| cartpole-swingup | chaos  | 7    | 856.5 | 861.5 | +0.59        | 1.78    | 2.15  |
| cartpole-swingup | chaos  | 15   | 859.2 | 860.6 | +0.16        | 1.81    | 2.18  |
| cartpole-swingup | clean  | 1    | 882.1 | 864.9 | -1.95        | 1.38    | 1.95  |
| cartpole-swingup | clean  | 7    | 878.4 | 882.4 | +0.45        | 1.37    | 1.62  |
| cartpole-swingup | clean  | 15   | 864.5 | 879.8 | +1.77        | 1.37    | 1.86  |
| cheetah-run      | chaos  | 1    | 354.5 | 501.1 | +41.35       | 1.94    | 2.36  |
| cheetah-run      | chaos  | 7    | 515.4 | 595.9 | +15.64       | 1.94    | 2.10  |
| cheetah-run      | chaos  | 15   | 528.2 | 540.9 | +2.41        | 1.95    | 2.19  |
| cheetah-run      | clean  | 1    | 926.2 | 946.6 | +2.20        | 1.51    | 1.62  |
| cheetah-run      | clean  | 7    | 944.0 | 942.7 | -0.14        | 1.50    | 1.64  |
| cheetah-run      | clean  | 15   | 850.7 | 905.6 | +6.45        | 1.50    | 1.63  |
| finger-turn-hard | chaos  | 1    | 651.0 | 448.7 | -31.08       | 1.95    | 2.05  |
| finger-turn-hard | chaos  | 7    | 580.5 | 838.8 | +44.50       | 1.94    | 2.21  |
| finger-turn-hard | chaos  | 15   | 510.1 | 753.5 | +47.71       | 1.97    | 2.06  |
| finger-turn-hard | clean  | 1    | 772.7 | 918.8 | +18.91       | 1.49    | 1.66  |
| finger-turn-hard | clean  | 7    | 818.2 | 917.0 | +12.07       | 1.50    | 1.60  |
| finger-turn-hard | clean  | 15   | 884.6 | 928.9 | +5.01        | 1.50    | 1.60  |
| hopper-hop       | chaos  | 1    | 166.8 | 278.1 | +66.71       | 2.15    | 2.46  |
| hopper-hop       | chaos  | 7    | 292.4 | 159.1 | -45.59       | 2.14    | 2.58  |
| hopper-hop       | chaos  | 15   | 158.2 | 139.0 | -12.17       | 2.17    | 2.55  |
| hopper-hop       | clean  | 1    | 542.8 | 570.1 | +5.04        | 1.72    | 1.83  |
| hopper-hop       | clean  | 7    | 514.9 | 469.7 | -8.78        | 1.71    | 1.86  |
| hopper-hop       | clean  | 15   | 559.3 | 357.9 | -36.00       | 1.73    | 1.87  |
| quadruped-run    | chaos  | 1    | 660.8 | 732.9 | +10.90       | 2.43    | 2.64  |

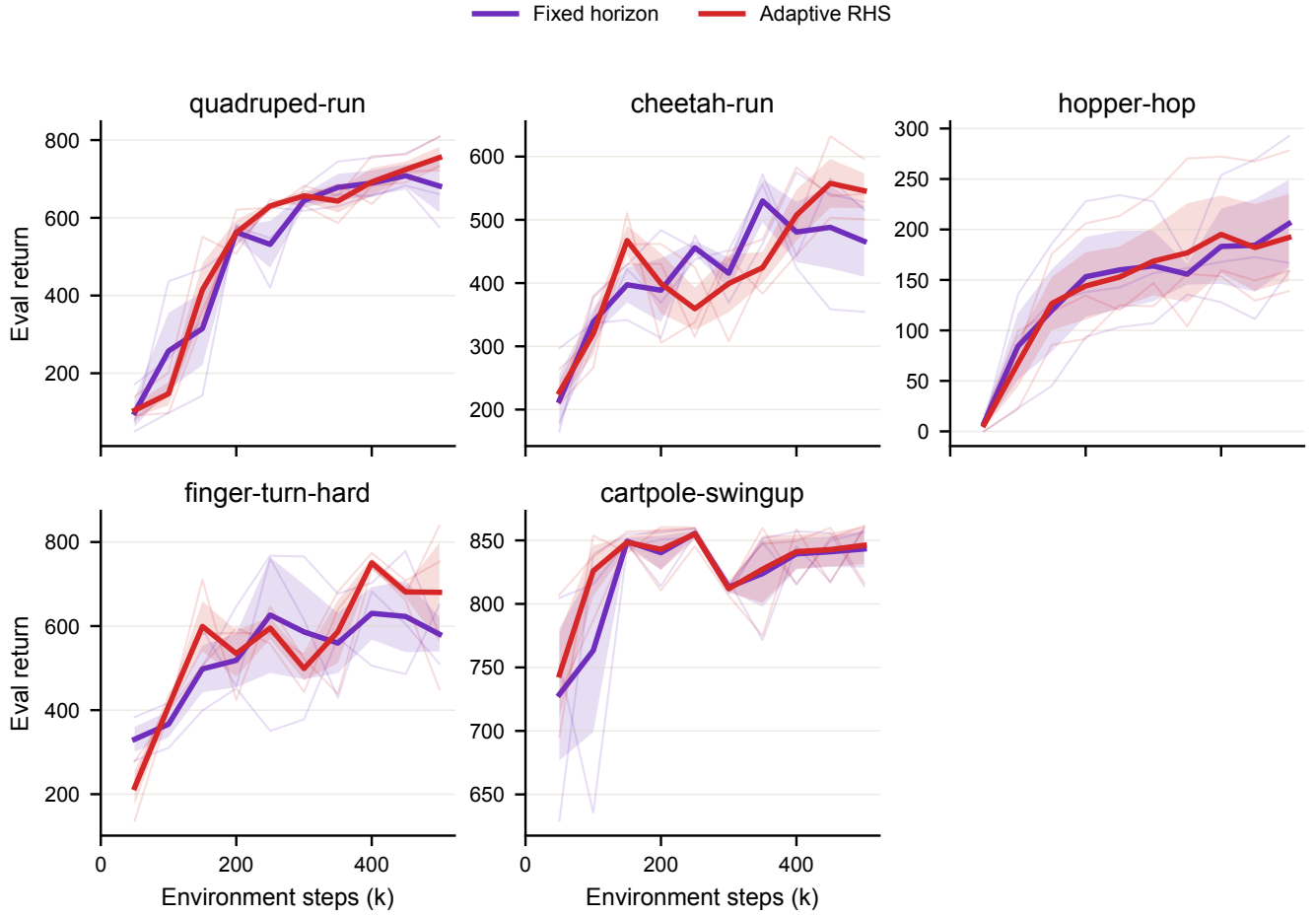
| Environment   | Regime | Seed | Fixed | RHS   | $\Delta$ (%) | Fixed h | RHS h |
|---------------|--------|------|-------|-------|--------------|---------|-------|
| quadruped-run | chaos  | 7    | 807.8 | 722.1 | -10.60       | 2.41    | 2.56  |
| quadruped-run | chaos  | 15   | 576.1 | 810.0 | +40.59       | 2.42    | 2.56  |
| quadruped-run | clean  | 1    | 881.8 | 913.6 | +3.60        | 1.93    | 2.13  |
| quadruped-run | clean  | 7    | 879.3 | 839.9 | -4.48        | 1.96    | 2.10  |
| quadruped-run | clean  | 15   | 911.0 | 923.8 | +1.41        | 1.99    | 2.25  |

## Learning Curves



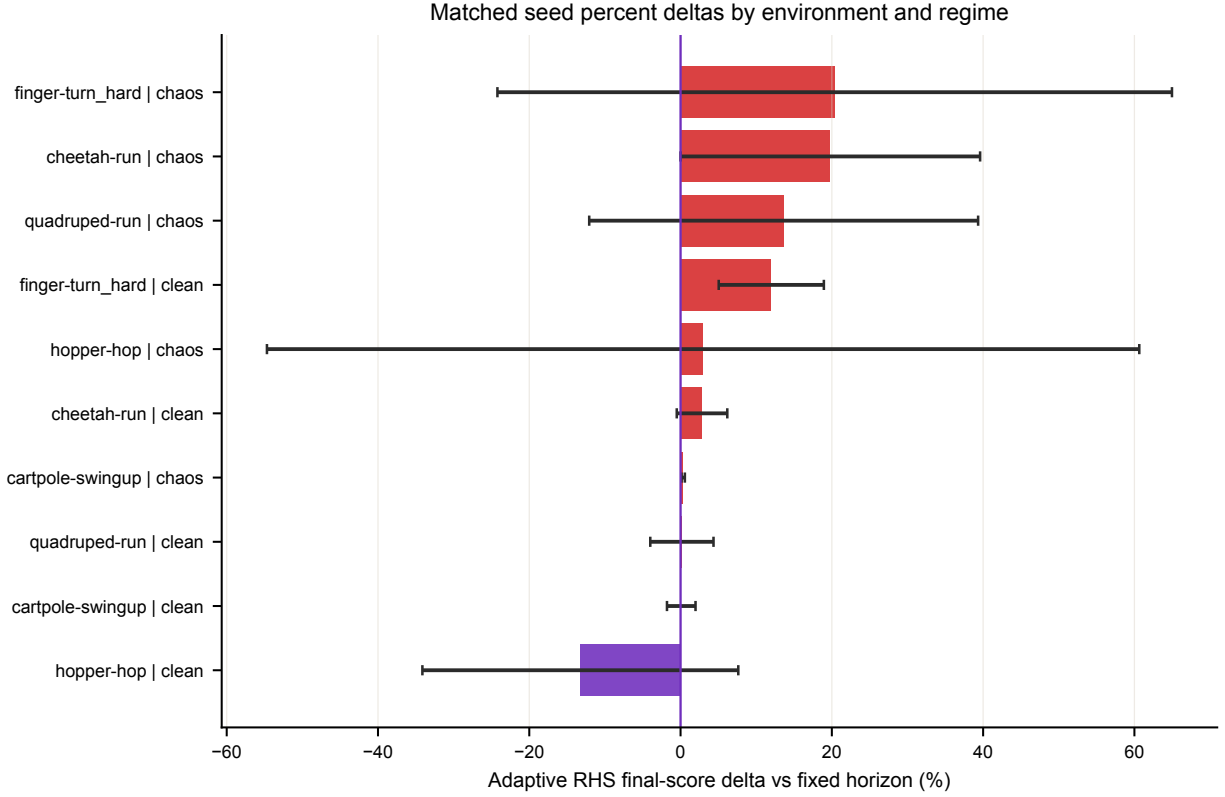
**Figure 1:** Clean-regime evaluation return. Thin traces are individual seeds; bold traces show the seed mean with standard error.

## Chaos regime learning curves



**Figure 2:** Chaos-regime evaluation return under matched randomized/noisy evaluation.

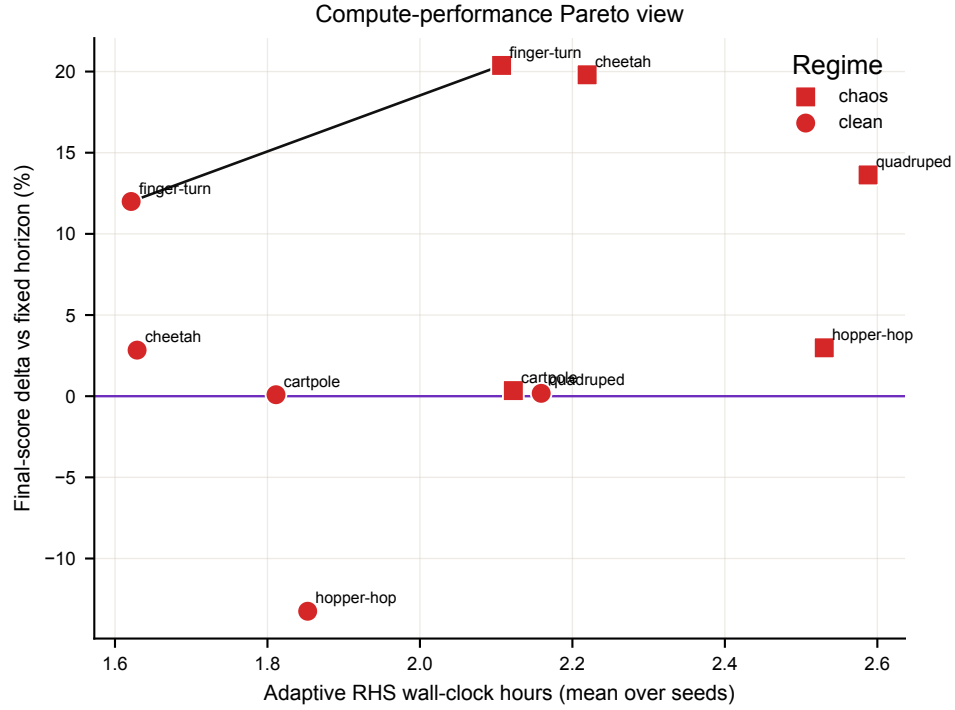
## Matched Percent Delta



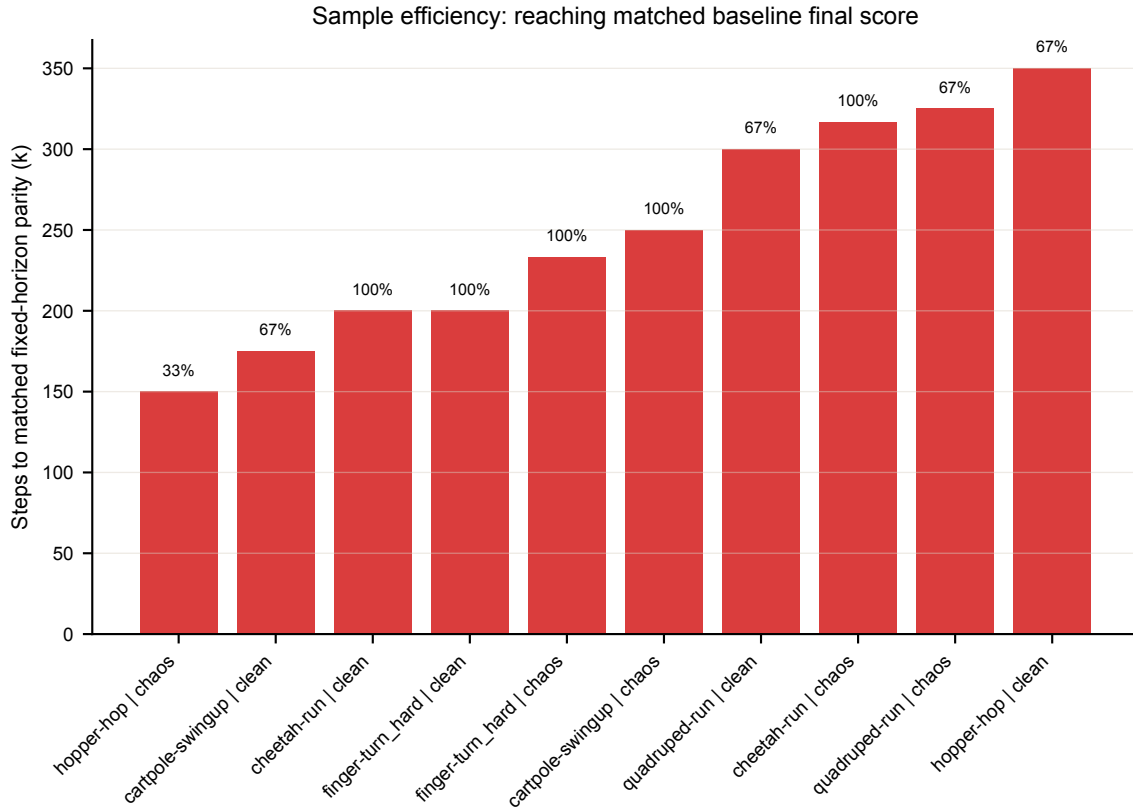
**Figure 3:** Per-environment and per-regime Adaptive RHS percent delta against the matched fixed-horizon baseline. Error bars are across seeds.

| Environment      | Regime | n | RHS delta (%)      | RHS wins | RHS h |
|------------------|--------|---|--------------------|----------|-------|
| cartpole-swingup | chaos  | 3 | $0.34 \pm 0.22$    | 100.0%   | 2.12  |
| cartpole-swingup | clean  | 3 | $0.09 \pm 1.89$    | 66.67%   | 1.81  |
| cheetah-run      | chaos  | 3 | $19.80 \pm 19.80$  | 100.0%   | 2.22  |
| cheetah-run      | clean  | 3 | $2.84 \pm 3.34$    | 66.67%   | 1.63  |
| finger-turn_hard | chaos  | 3 | $20.38 \pm 44.59$  | 66.67%   | 2.11  |
| finger-turn_hard | clean  | 3 | $11.99 \pm 6.95$   | 100.0%   | 1.62  |
| hopper-hop       | chaos  | 3 | $2.98 \pm 57.66$   | 33.33%   | 2.53  |
| hopper-hop       | clean  | 3 | $-13.25 \pm 20.88$ | 33.33%   | 1.85  |
| quadruped-run    | chaos  | 3 | $13.63 \pm 25.71$  | 66.67%   | 2.59  |
| quadruped-run    | clean  | 3 | $0.18 \pm 4.18$    | 66.67%   | 2.16  |

## Efficiency and Pareto View

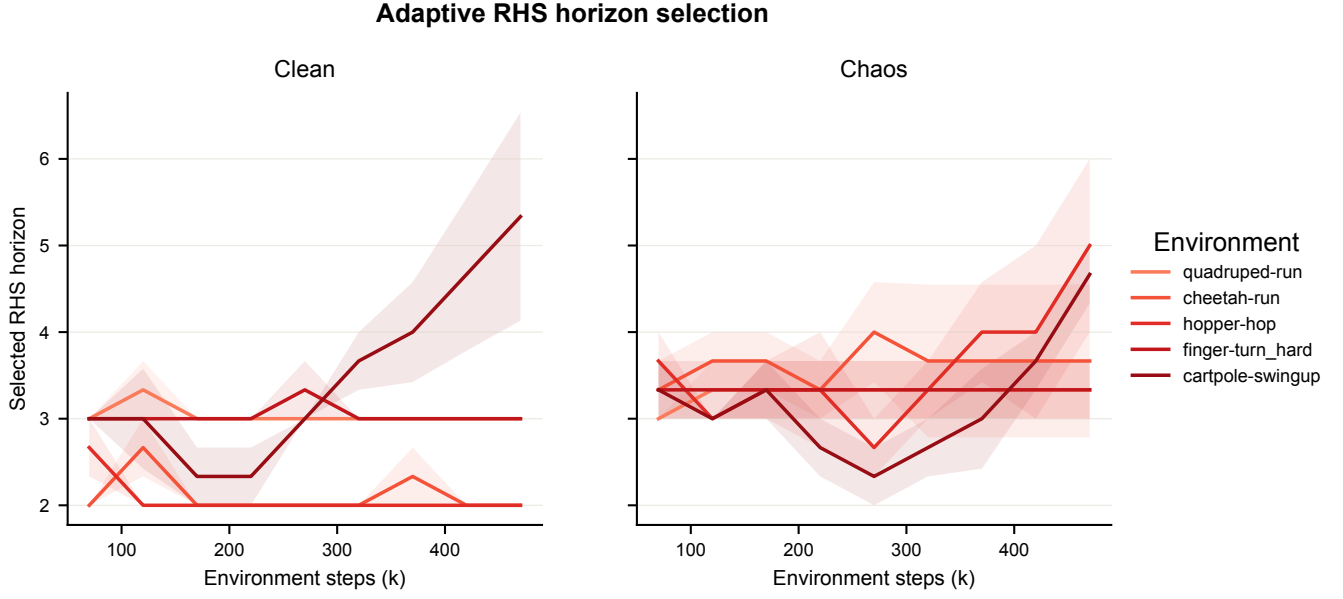


**Figure 4:** Compute-performance Pareto view. The x-axis is mean Adaptive RHS wall-clock hours and the y-axis is normalized final-score delta versus the matched fixed-horizon baseline; the black line highlights nondominated points.



**Figure 5:** Sample efficiency, measured as the first Adaptive RHS evaluation point that reaches the matched fixed-horizon final score. Labels above bars indicate the fraction of seeds reaching parity.

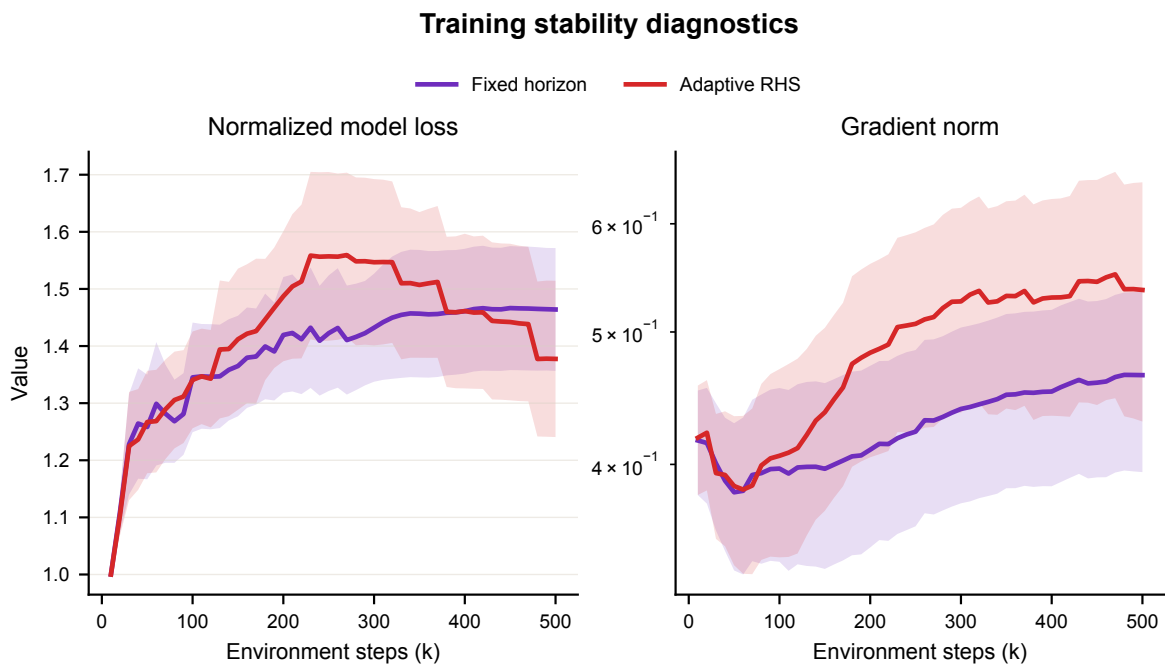
## RHS Diagnostics



**Figure 6:** Adaptive RHS selected horizon over training, averaged across seeds.

## Limitations

This is a preliminary report for the completed compact run after applying the report-level environment filter (60 retained terminal rows). Fish-swim is intentionally omitted because it was later removed from the v3 final matrix after isolated MJX chaos-push diagnostics exposed nonfinite state/reward failures. The final v3 package should reuse the same report template with the updated environment set and six seeds.



**Figure 7:** Training stability diagnostics pooled across runs. Total loss is normalized per run by its first finite value; gradient norms use a log axis.